

Impact of recombination mechanisms on the capacitance-voltage characteristics in bulk heterojunction organic solar cells

Durgesh C. Tripathi^{a,1}, Shailendra Kumar Gupta^{b,c,*}, Amit Kumar^c, Sandeep K. Pathak^c, Ashish Garg^d

^a The Zispel Nano Electronic Center, Department of Electrical Engineering, Technion-Israel Institute of Technology, Haifa 32000, Israel

^b Amity School of Physical Sciences, Amity University Punjab, IT City, Sector 82A, Mohali 140308, India

^c Department of Energy Science and Engineering, Indian Institute of Technology Delhi, Delhi 110016, India

^d Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, India

ARTICLE INFO

Keywords:

Capacitance-voltage
Negative capacitance
Drift-diffusion model
Langevin recombination
Trap assisted recombination

ABSTRACT

Low frequency Capacitance- voltage (C-V) measurements have been employed to investigate the effect of thickness variation of poly (3-hexylthiophene (P3HT) and fullerene [6,6],-phenyl-C₆₁-butyric acid methyl ester (PCBM) based Bulk Heterojunction (BHJ) photo absorbing layer in inverted organic solar cell (OSC) configuration, ITO|ZnO|P3HT:PCBM|MoO₃|Ag and are correlated with different recombination mechanisms. The normalized capacitance ($\frac{C}{C_{geom}}$) with voltage curves at different P3HT: PCBM blend thicknesses such as 130, 100, 80 and 70 nm demonstrate an increase in peak height with thickness as well as the presence of negative capacitance (NC) beyond peak voltage for higher thicknesses. To understand both peak height variation and the occurrence of NC, a defect sensitive simulation study has been performed by employing the radiative and non-radiative recombination processes. Based on the analysis of simulation results we confirm the role of Langevin bimolecular recombination rate in determining the peak height with thickness; however, the presence of NC in C-V curves can be understood through a competition between the Langevin bimolecular, and trap assisted recombination rates. Study confirms that peak height reduces due to increase in the rate of Langevin bimolecular recombination, whilst the trap assisted recombination resurges the NC beyond peak voltage. Later, the utilities of D-D based $\frac{1}{C^2}$ vs V analysis to calculate built-in voltage (V_{bi}) have been performed for the experimental and simulation data at different thicknesses emphasizing the strength of this approach to characterize threshold voltage of organic diodes.

1. Introduction

Organic photovoltaic (OPV) comprising of bulk heterojunction (BHJ) photo active layer have emerged as most promising strategy for photovoltaic applications in flexible or disposable electronics due to their low cost, light weight, mechanical flexibility and solution processability over large area devices [1–4]. The utilized organic semiconductors with low dielectric constant as absorbing materials, can easily be modified to obtain required absorption profile as well as electrical transport. Hence, significant improvement in the power conversion efficiency (PCE) has been recorded through synthesizing new materials, designing and engineering in new device architecture by the understanding of underlying

physical processes in OSC viz. the exciton generation, dissociation, transport and collection of carriers [1–5]. The active layer, based on BHJ, a composite of donor polymers (D) and acceptors (A) made up of poly (3-hexylthiophene) (P3HT) and fullerene [6,6],-phenyl-C₆₁-butyric acid methyl ester (PCBM) have been the subject of many investigations over the past decade. Besides PCBM as a component in the BHJ blend, a few nano-particles such as CdSe, ZnSe and PbSe have been proposed and their physics of charge and energy transfer has been established [6–8]. Under illumination, the generated excitons diffuse towards the P3HT-PCBM interfaces to dissociate into the free carriers and then these free charge carriers drift towards the respective electrodes in the presence of built-in voltage (V_{bi}) within the device due to work function

* Corresponding author at: Amity School of Physical Sciences, Amity University Punjab, IT City, Sector 82A, Mohali 140308, India.

E-mail addresses: shailendragupta5@gmail.com, skgupta@pb.amity.edu (S.K. Gupta).

¹ These authors contributed equally to the work.

difference between the electrodes [3,4,9].

Hence the processes of photo-generation, recombination and transport of charge carriers control the PCE performance and till now maximum ~16% PCE has been achieved for single junction OSC. The efficiency and long term stability of organic solar cells (OSC) compared to inorganic solar cells (Silicon, CGIS and CGTS) are still major impediments to their commercialization. Lesser PCE in OSC is primarily due to significant loss in exciton generation and charge generation due to various energy losses, radiative and non radiative recombination in the bulk as well as at the interfaces leading to current and voltage losses in the OSC devices. Besides, OSC often suffers from degradation due to the diffusion of molecular oxygen and moisture ingress into the device, degradation of photoactive layer and its reaction with electrodes [10]. However, such PCE is promising but still limited by charge carrier transport within the bulk as well as at the device interfaces due to presence of traps. It has been a general understanding that trap sites present at different locations in the device due to several different reasons play an important role in adversely impacting the PCE performance due to energy loss. The traps largely influence the exciton quenching, captures the free photo-generated carriers to decay internal electric field as well as get involved in trap assisted recombination causing the loss of mobility (μ) consequently reducing PCE [11–13]. The formation of these trapping states within the HOMO-LUMO (Highest Occupied Molecular Orbital – Lowest Unoccupied Molecular Orbital) levels of BHJ occurs due to positional and energetic disorders in the BHJ thin film during processing or with time and hence involved in several recombination processes impacting charge generation/transport within the device [3,4,14,15]. The deleterious role of charge carrier trapping has been shown to be very sensitive in controlling the performance of other organic devices in reducing the mobility as well as trap the free carriers [16–18].

To understand such various physical phenomenon related to traps, several characterization techniques have been employed to provide wide details regarding their role in charge transport [14–23] and recombination dynamics including transient techniques as well as current density-voltage (J - V) [24–26] and AC impedance spectroscopy (IS) [27–35]. To extract the frequency related electrical response of devices under traps, IS specifically capacitance spectra (capacitance – frequency and capacitance- voltage) of the organic diodes has been widely investigated where capacitance- voltage (C-V) curves are shown to be scanned from reverse to forward bias followed by primary regions as (i) constant capacitance in reverse bias due to the absence of inject carriers (ii) increased capacitance to a peak value (C_{peak}) (iii) decreased capacitance under higher forward bias beyond V_{peak} [36,37]. It has been reported that C_{peak} occurs near V_{bi} and the capacitance value approaches to the 3/4th of geometrical capacitance (C_{geom}) in the space charge limited (SCL) regime [28,32,38]. Though the Mott-Schottky (MS) relation has been commonly practiced to analyse the C-V curve prior to V_{peak} [27,31,39], tremendous efforts have been devoted to understand the underlying physics governing the occurrence of peak in the low-frequency C-V curves of organic diodes [28,32,35,38]. Mensfoort and Coehoorn proposed the involvement of diffusive and drift of carriers and suggested an empirical relation between V_{bi} and V_{peak} in the C-V curve for the determination of injection barriers [28]. Tripathi and Mohapatra demonstrated that the peak in the C-V curve is occurred due to competition between the diffusive and drift transport of injected carriers and demonstrated a relation between the diffusive capacitance and voltage that follows linear $\frac{1}{C^2}$ Vs V dependence in the diffusion dominated regime [32]. Recently, Tripathi et al. has investigated the impact of traps on the low frequency C-V characteristics of space charge limited diodes and demonstrated that the influence is significant in case of thicker active layer [23]. However in the literature, the utility of drift-diffusion (D-D) based expression to the BHJ solar cells has been illustrated and the relevant parameters such as V_{bi} and surface photo-voltage (V_{spv}) under illuminations have been successfully extracted [18,40,41].

Besides, the above-mentioned information, C-V curves provide sig-

nificant details on the role of traps present in the active layer and related interfaces in organic diodes primarily due to variations in active layer morphology [42], [43]. One such reason of obtaining morphological difference can be realized due to variation in active layer thickness [44]. During thickness variation device parameters such as reverse current and ideality factor play an important role in providing both qualitative and quantitative solutions regarding traps. In view of active layer (P3HT: PCBM) thickness variation, we have attempted to evaluate the impact of traps on P3HT: PCBM thickness sensitive C-V features under dark for ITO|ZnO|P3HT: PCBM|MoO₃|Ag configuration. Here, we have specifically chosen an inverted configuration due to its long-term stability and hence will not contain inherent traps due to unstable interface (normal OSC) and hence actual contribution of traps due to active layer thickness will be examined. The obtained C-V curves reveal a trend with thickness around V_{peak} . Besides peaks we also have observed the occurrence of negative capacitance (NC) beyond peak voltage (V_{peak}) under the forward bias operation specifically for thicker P3HT: PCBM already reported for organic diodes [33,38,45–50] attributing to different factors such as electron injection through the interfacial states [45], accumulation of charge carriers at heterojunction interface [51], self-heating [48,50], energetic disorder [38], Langevin bimolecular recombination [49], and trap-assisted monomolecular recombination [46,47] as well due to impurity concentration as reported by Quan Niu et al. [33] in the bipolar organic diodes. Hence, peak height variation with thickness as well as NC has also been evaluated in a numerical study based on drift- diffusion (D-D) model considering Langevin bimolecular or the trap assisted recombination mechanisms, that earlier is shown to affect the C-V curves with varying thicknesses which has not been widely investigated so far in the literature. Finally, the utility of C_{diff}^{-2} vs V analysis to determine the V_{bi} is examined for both the experimental and simulation C-V curves.

2. Experimental details

To fabricate the BHJ solar cells, poly (3-hexylthiophene) (P3HT) and phenyl-C61-butyric acid methyl ester (PCBM) were used as the photo-absorbing material system in the inverted device configuration, ITO|ZnO|P3HT: PCBM|MoO₃|Ag. In comparison to a normal OSC, in an inverted OSC device PEDOT: PSS is often replaced with ZnO (electron transport layer) to avoid unstable interface due to PEDOT: PSS moisture interaction. The incorporation of ZnO changes the polarity of device (flow of electrons towards ZnO/ITO) in comparison to a normal PEDOT: PSS (flow of holes towards PEDOT: PSS/ITO) and Ca/Al devices. MoO₃ is a hole transport layer between P3HT: PCBM and top electrode helps in blocking the flow of electrons and protects the P3HT: PCBM layer from thermally evaporated metal layer and hence ZnO and MoO₃ helps in providing long term stability in inverted OSC. The schematic architecture of in-house grown device is shown in Fig. 1(a). Initially, the pre-patterned ITO substrates were cleaned thoroughly in the boiled soap solution followed by rinsing in the DI water, then ultrasonicated in the acetone and IPA for 10 mins followed by drying using the pressurized N₂. The pre-cleaned ITO substrates were treated in the UV ozone for 20 mins prior to the ZnO layer deposition. The ZnO precursor solution was spin casted on the top of ITO at 2000 rpm followed by annealing at 250 °C for 10 min in the air. The ZnO coated substrates were transferred to the glove box (N₂ environment) for the P3HT: PCBM (1:0.8 wt. ratio in Chlorobenzene) processing with different spin speeds to get the variation in film thickness. P3HT: PCBM layers were thermally dried at 150 °C for 10 min to get the optimum morphology. The composition of blend and further annealing has been chosen according to our previous published work [52]. Thereafter, the substrates were loaded to thermal evaporator to deposit MoO₃/Ag top contact. The 10 nm of MoO₃ layer followed by 100 nm of Ag layer were evaporated sequentially at 0.5 Å/s deposition rate under high vacuum conditions. The finished devices with pixel area 9 mm² were taken out for electrical characterization

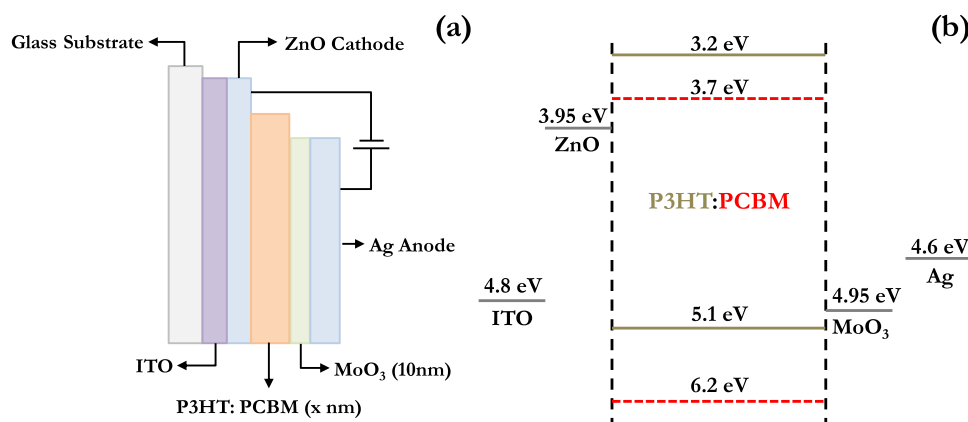


Fig. 1. Schematic of (a) an inverted organic BHJ solar cell device structure and (b) corresponding energy level diagram. Note that the thicknesses of photoactive layer (P3HT: PCBM) were varied from 70 nm to 130 nm by varying the spin speeds.

immediately after the electrode deposition. Detailed procedures of device fabrication have already been reported elsewhere [53,54]. The J-V characteristics of at least 4 devices for each thickness, under dark and illumination (AM 1.5 G) conditions were measured using a Newport solar simulator and the C-V measurements were performed using the Agilent 4294 A Precision Impedance Analyzer with an ac oscillating voltage of 50 mV (rms). The thickness of P3HT: PCBM layers was determined using the profilometer (Tencor Alpha-Step 500). Simulation of the C-V characteristics was performed using the Sentaurus technology computer-aided design (TCAD) platform.

3. Results and discussion

The OSC devices fabricated with different P3HT: PCBM thicknesses active layer in inverted configuration: ITO/ZnO/P3HT: PCBM (x nm)/MoO₃/Ag; device architecture and energy band diagram has been shown in Fig. 1(a-b), we reinvestigated under 1 sun (100 mWcm⁻²). The extracted parameters such as short circuit current (J_{sc}), open circuit voltage (V_{oc}), fill factor (FF) and PCE (%) are summarized in Table 1.

It is clear from the table, that the average J_{sc} and hence PCE increase with increase in P3HT: PCBM thickness from 70 to 130 nm due to enhanced absorption. The devices with 130 nm show highest PCE of 3.74% with maximum J_{sc} (10.70 mA/cm²). These devices were further used to measure IS spectroscopy (capacitance- voltage) which has been our prime motive behind this study.

Fig. 2 depicts the thickness dependent normalized capacitance ($\frac{C}{C_{geom}}$) vs voltage (V) plots measured under dark for all P3HT: PCBM thicknesses. Clearly $\frac{C}{C_{geom}}$ is found to be increased monotonically with increasing active layer thickness. However, for thick active layer devices such as 130 nm and 100 nm, we have observed the role of negative capacitance (NC) after the peak voltage under forward bias. Hence to understand increase in $\frac{C}{C_{geom}}$ and the occurrence of NC behavior we have opted 130 and 70 nm thicknesses to understand detailed comparative behavior observed in Fig. 2.

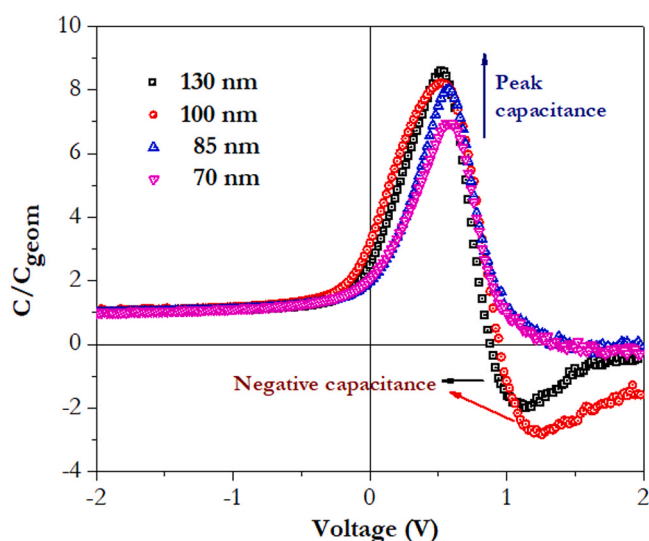


Fig. 2. C-V characteristics of organic BHJ solar cells under dark for 130, 100, 85 and 70 nm thick P3HT: PCBM layer.

3.1. Current density- voltage (J-V) characteristics

To explain C-V characteristics, discussed in previous section first we have explored the dark J-V for two different P3HT: PCBM thicknesses (70 and 130 nm) under dark and illuminated condition and is shown in Fig. 3. Evidently, the J-V characteristics are rectifying with more than two orders of magnitude of rectification ratio. The leakage current under reverse bias has been found to be thickness dependent and decreases substantially with increase in the P3HT: PCBM thickness. Various studies performed on bipolar organic diodes have reported the occurrence of such large leakage currents under reverse bias and different mechanisms have been proposed such as thermal activation of carriers at D-A interface [55,56], shunt path within the active layer [24,57], electrically and/or thermally activated carrier injection [25,58]. In this study, the expected barrier at anode and cathode electrodes is quite high for opposite charge carrier injection and hence, electrically and/or thermally activated carrier injection may not be possible. Furthermore, suppression of leakage current with increasing thickness indicates the dominant mechanisms are very sensitive to the thicknesses.

Under forward bias operation, the current increases exponentially over three orders of magnitude and eventually enters the SCL regime. The magnitude of current in the exponential regime is found to be thickness independent; however, it is slightly higher for thicker P3HT:

Table 1

Photovoltaic performance of OSCs devices based on ITO/ZnO/xP3HT: PCBM/MoO₃/Ag structure with varying active layer thicknesses.

Thickness (nm)	J_{sc} (mAcm ⁻²)	V_{oc} (Volt)	FF	R_s (Ωcm ²)	R_{sh} (Ωcm ²)	PCE (%)
130	10.7	0.60	0.58	10.0	625	3.74
100	9.67	0.65	0.58	11.2	585	3.64
85	9.46	0.65	0.58	10.7	714	3.56
70	9.31	0.65	0.58	10.0	652	3.52

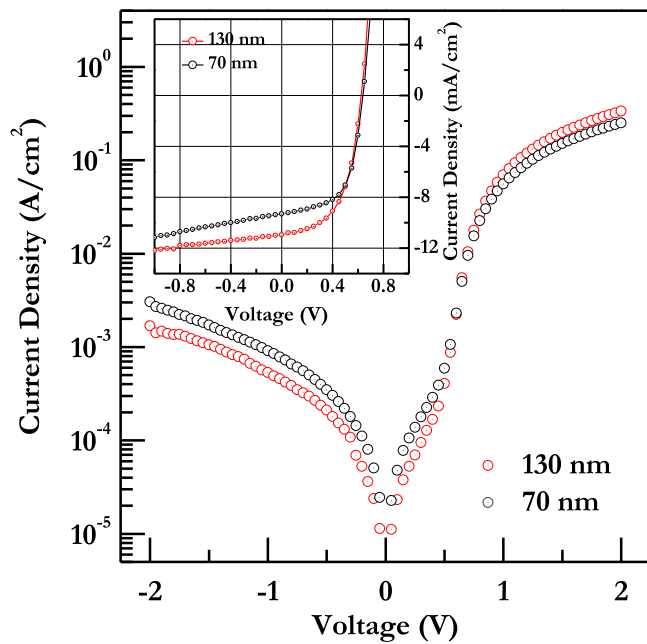


Fig. 3. J-V characteristics of organic BHJ solar cells under dark for 130 and 70 nm thick P3HT: PCBM layer. Inset shows the J-V characteristics under 100 mW/cm² illumination, for two thicknesses. Note that the magnitude of currents in the SCL regime is higher for thicker devices.

PCBM. The SCL current in organic diodes is usually understood by the Mott Gurney relationship, $J = \frac{9}{8} \mu \epsilon \frac{V^2}{d^3}$, under the field independent mobility condition [59]. Here, V represents the externally applied voltage, μ is the charge carrier mobility, ϵ is the permittivity, and d is the active layer thickness. This expression predicts that the SCL current should scale with d^{-3} , i.e. higher SCL current for lower P3HT: PCBM thickness. It is believed that the thickness variation in P3HT: PCBM does not change the injection efficiency of an electrode and the charge carrier mobility. Hence, observation of higher current in the thicker devices must be associated with carrier recombination process which is not considered in the Mott Gurney relationship.

The extracted parameters under dark and illuminated J-V characteristics for specific 130 and 70 nm thick devices have been tabulated in Table 1. The PCE values are found to be 3.75 and 3.50 for 130 nm and 70 nm thick P3HT: PCBM devices, respectively, which are found to be similar to reported values for the same P3HT: PCBM polymer blend system [3,4] and hence, adequate to carry out further study.

3.2. Capacitance - voltage (C-V) characteristics

The C-V characteristics, measured at 1 kHz frequency, for 130 and 70 nm diodes have been depicted in Fig. 4 along with $\frac{C}{C_{geom}}$ Vs V in inset. Evidently, the capacitance remains constant under reverse bias operation and corresponds to P3HT: PCBM thickness. In the forward bias, a

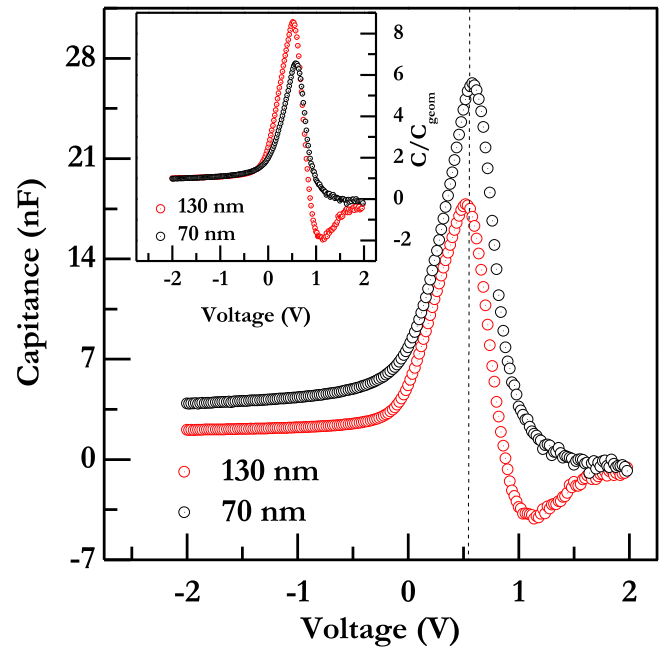


Fig. 4. C-V characteristics of organic BHJ solar cells measured at 1 kHz for 130 and 70 nm thick P3HT: PCBM layer. Inset shows the C-V curve normalized with reverse bias capacitance for both thicknesses at same frequency. Note that the C-V peak voltage does not change with photoactive layer thickness.

prominent peak is observed around 0.6 V and this peak voltage (V_{peak}) is found to be independent of P3HT: PCBM thicknesses. It should be noted that the devices with varying P3HT: PCBM thicknesses were also fabricated, and they all exhibited similar C-V characteristics in terms V_{peak} with thickness variation (Fig. 2). It has already been reported that the occurrence of peak in the C-V curve is associated with competition between the diffusion and drift component of injected carriers at the beginning of SCL transport [32]. In the present study, V_{peak} is found to be very close to the theoretical V_{bi} , shows good agreement with the previous study [32].

As shown in the inset $\frac{C}{C_{geom}}$ Vs V curve, peak height exhibits thickness dependence and is equal to $6.5C_{geom}$ and $8.5C_{geom}$ for the 70 nm and 130 nm thick OSC devices; however, V_{peak} remains unchanged. Earlier studies performed on single carrier organic diodes demonstrated that the energetics of injecting electrode determine the peak height and is found to be $1.405C_{geom}$ with no injection barrier at one of the electrodes [28]. Observation of very intense C-V peak indeed manifests the importance of bulk charge carrier processes, such as recombination, in addition to electrode energetic in case of bipolar organic diodes. Furthermore, at large forward bias above V_{peak} , the capacitance becomes negative (observation of NC) for 130 nm thick OSC devices. To get more insight, we have further performed a simulation study to elucidate the observed features in the C-V characteristics by considering the carrier recombination.

3.3. TCAD Simulation

Here we have performed a simulation study on TCAD platform in order to develop a detailed physical understanding on the impact of carrier recombination mechanisms (radiative and non-radiative) on low-frequency C-V characteristics [60]. In our model, the energy levels of P3HT: PCBM blend are defined by considering the A_{LUMO} (Acceptor LUMO) and D_{HOMO} (Donor HOMO). The mobility (μ) of injected carriers is assumed to follow the standard Poole-Frenkel type of electric field dependence [$\mu(E, T) = \mu_0(T) \exp(\beta E^{1/2})$], i.e. carrier mobility increases exponentially with square root of the electric field [61], with following parameters $\mu_{o,h}(T) = 10^{-3} \text{ cm}^2/\text{V-s}$, $\mu_{o,e}(T) = 10^{-3} \text{ cm}^2/\text{V-s}$, $\beta_{o,h}(T) =$

Table 2

The parameters derived from J-V characteristics measured under dark and illumination for two photoactive layer thicknesses.

Parameter	130 nm	70 nm
V_{oc}	0.65	0.65
J_{sc}	10.96 mA/cm ²	9.31 mA/cm ²
FF	0.58	0.58
Power Conversion Efficiency	3.75	3.50
R_s (from J-V at one sun)	10.5 Ω cm ²	11.8 Ω cm ²
R_{sh} (from J-V at one sun)	570 Ω cm ²	694 Ω cm ²
Ideality Factor (from Dark J-V)	2.2	2.9
Rectification Ratio (from Dark J-V)	300	100

$4 \times 10^{-3}(\text{cm/V})^{1/2}$, and $\beta_{o,e}(T) = 7 \times 10^{-3}(\text{cm/V})^{1/2}$ at room temperature. For the sake of simplicity, the carrier concentration dependent carrier mobility is not considered. Additionally, the electron trap states with typical concentration (N_t) $1 \times 10^{16} \text{ cm}^{-3}$ have been considered at mid-gap level to facilitate the trap assisted recombination. The values of all relevant parameters taken in this simulation study are summarized in Table 3.

The direct recombination of free carriers in bipolar diodes typically obeys Langevin bimolecular recombination rate (R_{BM}) which is given by

$$R_{BM} = \gamma \frac{q}{\epsilon} (\mu_n + \mu_p) np$$

where q is the elementary charge, ϵ is the permittivity of P3HT:PCBM, μ_n and μ_p are the electron and hole mobility, n and p are the free electrons and holes concentration, and γ is the dimensionless Langevin reduction factor. Experimentally observed value of γ is found to be in the range of 0.5–0.001 for organic materials [62]. The Langevin bimolecular recombination rate (R_{BM}) can be varied by varying the Langevin reduction factor (γ) and hence the lifetime (τ_r) of free carriers. This approach has been adopted here to vary the recombination rate to investigate the effect on low-frequency C-V characteristics.

The electron traps commonly found in the organic semiconductors not only limit the electron transport, but also facilitate the carrier recombination between the trapped electrons and free holes through the non-radiative trap assisted recombination process. The trap assisted recombination rate (R_{SRH}) is given by

$$R_{SRH} = \frac{C_n C_p N_t (np - n_1 p_1)}{[C_n(n + n_1) + C_p(p + p_1)]}$$

Where, N_t is the electron trap concentration, C_n and C_p are the electron and hole capture coefficient, and $n_1 p_1 = N_{cv} \exp[-E_{gap}/kT] = n_i^2$ their product under equilibrium condition [14]. The capture coefficient is related to mobility by $(q/\epsilon_0 \epsilon_r) \mu_{n,p}$. For the trapped assisted recombination between the free holes and trapped electrons, the above expression can be simplified using, $C_n = C_p$, $np \gg n_1 p_1$, and $n \approx p$ to

$$R_{SRH} \approx \frac{C_p N_t p}{2}$$

The above expression suggests that the rate of trap assisted recombination (R_{SRH}) increases linearly either with the carrier capture coefficient or trap concentration (C_n/p). The lifetime (τ) of trap assisted recombination in bipolar diodes is then approximated as $\tau_r = 2/N_t C_p$. In this study, the value of C_p is kept constant as $1.56 \times 10^{-12} \text{ cm}^3/\text{s}$ and the trap assisted recombination rate (R_{SRH}) is varied by varying the trap concentrations (N_t).

3.4. Simulated C-V Characteristics

First, we have attempted a detailed comparison between the normalized simulated C-V curves of unipolar and bipolar diodes having similar V_{bi} (equal to 1.0 V). In doing this, the LUMO level of P3HT:PCBM

Table 3
Parameters used for Numerical Simulations.

Parameters	Value
Thickness of photoactive layer	70 nm/130 nm
Dielectric Permittivity of Photoactive layer	3.9
Bandgap	1.4 eV
LUMO/HOMO Level	3.7/5.1 eV
Effective Density of states LUMO (HOMO)	$5 \times 10^{20} \text{ cm}^{-3}$
Mobility of electrons and holes	$1 \times 10^{-3} \text{ cm}^2/\text{V-s}$
Work function of Anode/Cathode	4.95/3.95 eV
Bimolecular recombination coefficient (varied)	0.35–0.01
Trap Concentration (varied)	$1-10 \times 10^{16} \text{ cm}^{-3}$
Trap Level	Mid-gap

in the unipolar diode has been shifted to 3.1 eV to suppress any.

possibility of electron injection from the cathode i.e. only holes carrier could contribute to the total current. Simulated C-V curves of both unipolar and bipolar diode structures for 70 nm active layer thickness is shown in Fig. 5. A peak in C-V curves is observed near to theoretical V_{bi} for both structures. Indeed, the peak height of C-V curve has the impression of carrier injections and the capacitance features beyond the V_{peak} in forward direction is noticeably different for both structures. As in the case of unipolar diode, the peak height is approximately $1.30C_{geom}$ and approaches to the drift-only value $0.75C_{geom}$ at large voltage limit showing agreement with Mensfoort and Coheroon study [28]. Furthermore, the peak capacitance (C_{peak}) of bipolar diode is found to be intense, $1.50C_{geom}$, and the capacitance beyond the V_{peak} shows non-monotonic voltage dependence. In unipolar diode, the peak capacitance may also become intense particularly in presence of traps however the capacitance value beyond the peak voltage never become negative [23]. Occurrence of negative capacitance in this study demonstrate the importance of carrier recombination in determining the C-V characteristics of bipolar diodes. In the following, objective of this study is to investigate how the different recombination processes control the low frequency C-V characteristics.

Fig. 6(a-b) shows the simulated capacitance curves of 70 nm thick bipolar diodes for varying Langevin bimolecular, and trap assisted recombination rates, respectively. In doing this, the trap concentration (N_t) was kept constant at $1 \times 10^{16} \text{ cm}^{-3}$ in case of bimolecular recombination rate variation and the Langevin reduction factor (γ) was kept at 0.1 for the case of trap assisted recombination rate variation. Evidently, the bimolecular recombination rate changes the C-V characteristics around the V_{peak} as shown in Fig. 6(a). The C-V peak height is found to increase as bimolecular recombination rate (R_{BM}) decreases. Simultaneously, the capacitance beyond the V_{peak} also becomes more negative due to dominance of trap assisted recombination over the Langevin bimolecular recombination. These observations indeed suggest that the reduced bimolecular recombination rate surges the C-V peak height as well as negative capacitance beyond the V_{peak} .

The C-V characteristics for varying trap concentration (N_t) at fixed

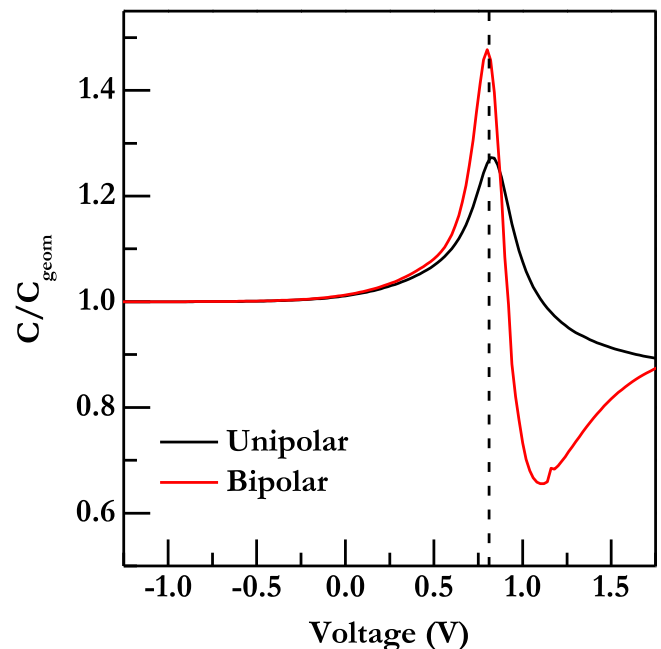


Fig. 5. Simulated capacitance curve for unipolar and bipolar diodes with 70 nm active layer thickness. The value of Langevin reduction factor (γ) in the bipolar diode is taken 0.35. Note that the C-V peak voltage (V_{peak}) is almost same for both diodes however, the peak height clearly has impression of charge injection.

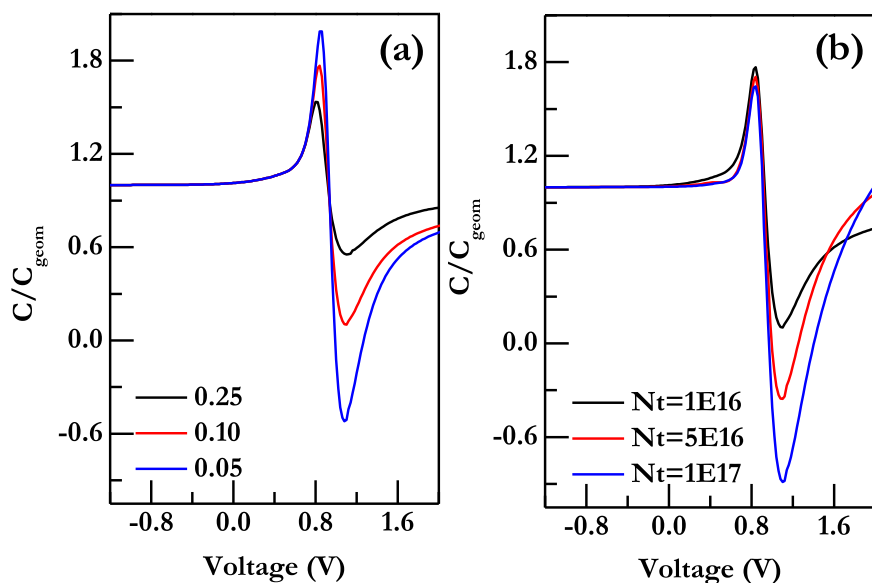


Fig. 6. Simulated C-V curves for two different variation (a) Langevin reduction factor (γ) and (b) trap concentration (N_t). Note that the electron trap concentration is kept constant at $1 \times 10^{16} \text{ cm}^{-3}$ for Langevin reduction factor variation while the electron trap concentrations are varied keeping Langevin reduction factor constant at 0.1 in the second case. The thickness of photoactive layer is 70 nm in both cases.

Langevin reduction factor (γ) are depicted in Fig. 6(b). Clearly, in this case, the C-V peak height is reduced slightly, whereas the magnitude of negative capacitance is increased significantly with increasing trap concentration (N_t). As expected, increasing the trap concentration enhances the trap assisted recombination rate as a result; the recombination of trapped electrons with free holes is dominated leading to increase in the magnitude of negative capacitance as can be seen in Fig. 6(b). Our simulation study shows good agreement with Niu et al. [33], performed for the polymeric light emitting diodes. Our study further manifests that the low-frequency C-V characteristics have a distinguished feature beyond the peak voltage signify dominance of different carrier recombination mechanisms in the organic bipolar diodes.

Based on the results of above simulated C-V studies, we have attempted to strengthen our understanding for the C-V characteristics displayed in Fig. 4. In Fig. 7, we simulated C-V curves for two different thicknesses (130 and 70 nm) as these thicknesses show clear trend in normal capacitance as well as the occurrence of negative capacitance by considering both recombination processes. First, the value of Langevin reduction factor (γ) is reduced slightly keeping the trap concentrations constant for 130 nm thick active layer device to generate the features of experimental curves qualitatively. Indeed, with reducing the value of γ , the C-V peak height and the magnitude of negative capacitance are increased significantly and demonstrated the experimental C-V curve features qualitatively. The study suggests that the surge of peak height in experimental data can be attributed to the reduction in the bimolecular Langevin recombination rate. In literature, several studies have reported the reduced bimolecular Langevin recombination rate (R_{BM}) usually one to four orders of magnitude in the low mobility organic solids [20, 62–64]. Different proposals have been made to explain this finding including the potential fluctuation [63], determined by the slowest charge carrier [20], and the concentration gradient within the devices [62]. Here, we attribute the variation in morphology of thin films due to phase separation of donor and acceptor molecules may be responsible for reduced bimolecular Langevin recombination rate for thick active layer device in the present case. This study opens an opportunity for detailed investigation of low-frequency C-V curve under illumination to understand the recombination mechanisms.

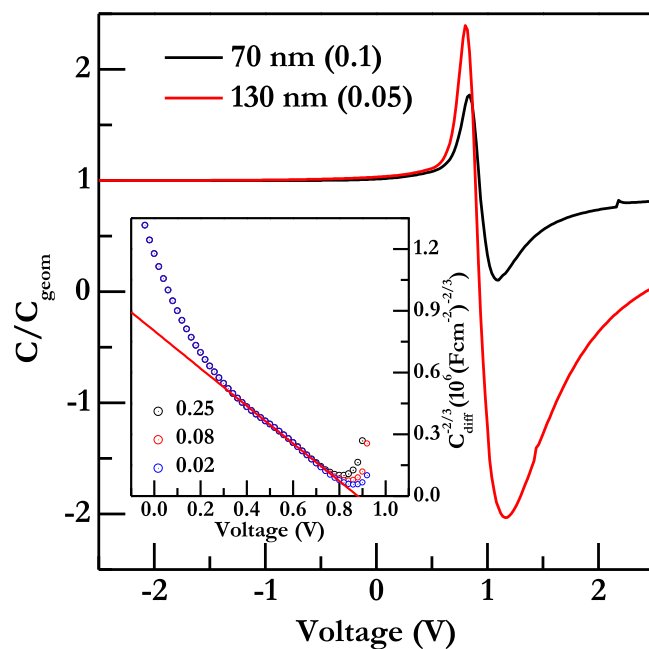


Fig. 7. Simulated C-V curves for two different active layer thicknesses (70 nm and 130 nm) considering Langevin bimolecular and trap assisted recombination processes. The value of trap concentration is kept $1 \times 10^{16} \text{ cm}^{-3}$ and Langevin reduction factor is reduced to produce qualitatively the experimental curves displayed in the Fig. 4. The value of Langevin reduction factor is given in the bracket. The inset shows the $C_{diff}^{-2/3}$ vs V analysis of C-V curves for 70 nm thick bipolar diode with varying the Langevin bimolecular recombination rates.

3.5. $C^{-2/3}$ vs V Analysis

Following, we present an analysis on the rise of capacitance prior to the peak voltage (V_{peak}) for both experimental and simulated C-V curves. It is demonstrated that the diffusive capacitance (C_{diff}) prior to V_{peak} originated due to the storage of additional carriers under the carrier profile leads to a linear voltage, $C_{diff}^{-2/3} \propto (V - V_{bi})$, dependence [32]. The

C_{diff} from the measured capacitance is obtained by subtracting the geometrical capacitance (C_{geo}). In the inset of Fig. 6, $C_{\text{diff}}^{-2/3}$ is plotted for varying Langevin bimolecular recombination rate (R_{BM}) in the diffusive dominated regime. Evidently, the variation of recombination rate does not affect the linear regime and hence, all the curves can be fitted excellently with a straight line. The voltage axis intercepts of fitted line at 0.88 V which is slightly lower than the theoretical $V_{\text{bi}} = 1.0$ V, and relatively higher than the $V_{\text{peak}} = 0.78$ V. The difference between the intercept voltage and theoretical V_{bi} is 0.12, which could be associated with diffusion induced band bending at both electrodes. The intercept voltage is inferred as a threshold for beginning of SCL injection with drift of carriers dominating over the diffusion as reported in the previous study [32,40].

Experimental C-V curve of different thicknesses are also plotted in the $C_{\text{diff}}^{-2/3}$ vs V fashion as shown in Fig. 8. Indeed, a linear regime can be marked for both thicknesses in the diffusion dominated regime of $C_{\text{diff}}^{-2/3}$ vs V plots. The linear regime of both curves is fitted excellently with straight lines. Though, the slope of fittings is different for different thicknesses, the intercepts on the voltage axis is same, 1.05 V, for both P3HT:PCBM thicknesses. In contrast to MS analysis, which estimates the thickness dependent threshold voltage [29], the $C_{\text{diff}}^{-2/3}$ analysis is robust and can be used routinely to investigate the threshold voltage in the organic diodes.

4. Conclusions

In conclusion, we have extensively studied the low-frequency C-V behavior of P3HT:PCBM based BHJ solar cells for 130 and 70 nm thicknesses in the inverted OSC configuration. It has been observed that the voltage corresponding to the C-V peak remains intact, however the peak height is significantly changed. Besides the occurrence of NC for 130 nm active layer beyond the P_{peak} has been established. The performed simulation studies based on D-D model suggests the role of Langevin bimolecular recombination in diminishing the peak height and the NC beyond the V_{peak} . Besides, enhancement of trap assisted recombination by adding impurity concentration reduces the C-V peak height slightly however, the magnitude of negative capacitance increases substantially. This study further suggests that the low-frequency C-V curves have distinguished aspects of different recombination mechanisms beyond the peak voltage in the forward direction. The $C_{\text{diff}}^{-2/3}$ vs. V analysis, originally developed for the single carrier SCL diodes, is adequate even for bipolar diodes to investigate the threshold voltage from the C-V curves. This study further opens more opportunities to quantify the trap densities in the organic bipolar devices utilizing the low-frequency C-V curve into the NC region.

CRediT authorship contribution statement

Durgesh C Tripathi: Conceptualization, Methodology, Software, Validation. **Shailendra Kumar Gupta:** Conceptualization, Methodology, Writing – original draft. **Amit Kumar:** Writing – review & editing. **Sandeep K. Pathak:** Writing – review & editing, Supervision. **Ashish Garg:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shailendra Kumar Gupta reports financial support was provided by Science and Engineering Research Board. Shailendra Kumar Gupta reports a relationship with Science and Engineering Research Board that includes: funding grants. The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

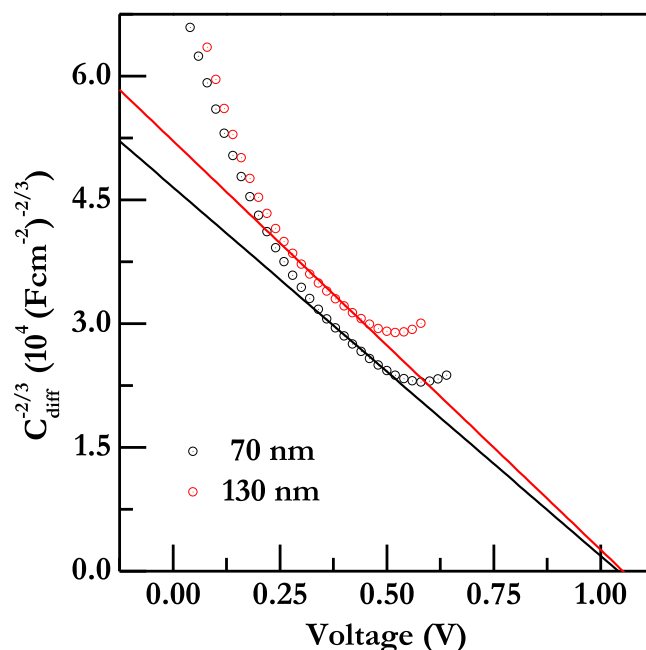


Fig. 8. The, vs V analysis of experimental C-V curves of BHJ solar cells for 130 and 70 nm of P3HT:PCBM thicknesses. Note that the voltage axis intercepts of fitted straight line is same for both thicknesses.

Data Availability

The authors do not have permission to share data.

Acknowledgments

The authors thank the Science and Engineering Research Board (SERB), Department of Science and Technology, Government of India (DST) for financial support. The work has been completed under the N-PDF (National post doctorate fellowship) project PDF/2017/002334.

Author Contributions

D.C.T. and S.K.G. contributed equally to this work. All authors have given approval to the final version of the manuscript.

References

- [1] R.S. Gurney, D.G. Lidzey, T. Wang, A review of non-fullerene polymer solar cells: from device physics to morphology control, Rep. Prog. Phys. 82 (2019), <https://doi.org/10.1088/1361-6633/ab0530>.
- [2] G.J. Hedley, A. Ruseckas, I.D.W. Samuel, Light harvesting for organic photovoltaics, Chem. Rev. 117 (2017) 796–837, <https://doi.org/10.1021/acs.chemrev.6b00215>.
- [3] L. Lu, T. Zheng, Q. Wu, A.M. Schneider, D. Zhao, L. Yu, Recent advances in bulk heterojunction polymer solar cells, Chem. Rev. 115 (2015) 12666–12731, <https://doi.org/10.1021/acs.chemrev.5b00098>.
- [4] A.J. Heeger, 25th anniversary article: bulk heterojunction solar cells: understanding the mechanism of operation, Adv. Mater. 26 (2014) 10–28, <https://doi.org/10.1002/adma.201304373>.
- [5] Y. Cui, H. Yao, J. Zhang, T. Zhang, Y. Wang, L. Hong, K. Xian, B. Xu, S. Zhang, J. Peng, Z. Wei, F. Gao, J. Hou, Over 16% efficiency organic photovoltaic cells enabled by a chlorinated acceptor with increased open-circuit voltages, Nat. Commun. 10 (2019) 2515, <https://doi.org/10.1038/s41467-019-10351-5>.
- [6] N. Mastour, Z. Ben Hamed, A. Benchaabane, M.A. Sanhoury, F. Kouki, Effect of ZnSe quantum dot concentration on the fluorescence enhancement of polymer P3HT film, Org. Electron. 14 (2013) 2093–2100, <https://doi.org/10.1016/j.orgel.2013.05.005>.
- [7] N. Mastour, H. Bouchriha, Effect of CdSe nanoparticles on the fluorescence spectra of conjugate polymer P3HT: an experimental and theoretical study, Phys. Lett. A 380 (2016) 3866–3869, <https://doi.org/10.1016/j.physleta.2016.09.047>.
- [8] N. Mastour, R. Ridene, H. Bouchriha, Charge and non-radiative energy transfer in P3HT-CdSe nanocomposite, Int. J. Hydrog. Energy 42 (2017) 8813–8817, <https://doi.org/10.1016/j.ijhydene.2016.10.138>.

- [9] P.K. Manda, S. Ramaswamy, S. Dutta, Extraction of the built-in potential for organic solar cells from current-voltage characteristics, *IEEE Trans. Electron Devices* 65 (2018) 184–190, <https://doi.org/10.1109/TED.2017.2773708>.
- [10] V.I. Madogni, B. Kounouhewa, A. Akpo, M. Agbomahewa, S.A. Hounkpatin, C. N. Awanou, Comparison of degradation mechanisms in organic photovoltaic devices upon exposure to a temperate and a subequatorial climate, *Chem. Phys. Lett.* 640 (2015) 201–214, <https://doi.org/10.1016/j.cplett.2015.09.023>.
- [11] S. Zeiske, O.J. Sandberg, N. Zarrabi, W. Li, P. Meredith, A. Armin, Direct observation of trap-assisted recombination in organic photovoltaic devices, *Nat. Commun.* 12 (2021) 3603, <https://doi.org/10.1038/s41467-021-23870-x>.
- [12] Z. Chen, T. Wang, Z. Wen, P. Lu, W. Qin, H. Yin, X.-T. Hao, Trap state induced recombination effects on indoor organic photovoltaic cells, *ACS Energy Lett.* 6 (2021) 3203–3211, <https://doi.org/10.1021/acseenergylett.1c01336>.
- [13] R. Jasiuⁿas, V. Jasinskas, H. Zhang, T. Upreti, F. Gao, M. Kemerink, V. Gulbinas, Carrier mobility dynamics under actual working conditions of organic solar cells, *J. Phys. Chem. C* 125 (2021) 14567–14575, <https://doi.org/10.1021/acs.jpcc.1c04245>.
- [14] M. Kuik, G.-J.A.H. Wetzelaer, H.T. Nicolai, N.I. Craciun, D.M. De Leeuw, P.W. M. Blom, 25th anniversary article: charge transport and recombination in polymer light-emitting diodes, *Adv. Mater.* 26 (2014) 512–531, <https://doi.org/10.1002/adma.201303393>.
- [15] Y. Zhang, X. Li, T. Dai, D. Xu, J. Xi, X. Chen, Charge transport and extraction of PTB7:PC 71 BM organic solar cells: effect of film thickness and thermal-annealing, *RSC Adv.* 9 (2019) 24895–24903, <https://doi.org/10.1039/C9RA02877C>.
- [16] G. Zuo, M. Linares, T. Upreti, M. Kemerink, General rule for the energy of water-induced traps in organic semiconductors, *Nat. Mater.* 18 (2019) 588–593, <https://doi.org/10.1038/s41563-019-0347-y>.
- [17] K.S. Rao, D.C. Tripathi, Y.N. Mohapatra, Carrier capture kinetics at electrical defects in poly [2-methoxy-5-(2-ethyl-hexyloxy)-1,4-phenylenevinylene] (MEH-PPV), *Stud. Using Charge. transient Spectrosc.*, *J. Appl. Phys.* 116 (2014), 054511, <https://doi.org/10.1063/1.4891832>.
- [18] U.K. Verma, D.C. Tripathi, Y.N. Mohapatra, Direct determination of defect density of states in organic bulk heterojunction solar cells, *Appl. Phys. Lett.* 109 (2016), 113301, <https://doi.org/10.1063/1.4962827>.
- [19] K. Harada, A.G. Werner, M. Pfeiffer, C.J. Bloom, C.M. Elliott, K. Leo, Organic homojunction diodes with a high built-in potential: interpretation of the current-voltage characteristics by a generalized einstein relation, *Phys. Rev. Lett.* 94 (2005), 036601, <https://doi.org/10.1103/PhysRevLett.94.036601>.
- [20] L.J.A. Koster, V.D. Mihailescu, P.W.M. Blom, Bimolecular recombination in polymer/fullerene bulk heterojunction solar cells, *Appl. Phys. Lett.* 88 (2006) 1–3, <https://doi.org/10.1063/1.2170424>.
- [21] N.C. Giebink, G.P. Wiederrecht, M.R. Wasielewski, S.R. Forrest, Ideal diode equation for organic heterojunctions. I. Derivation and application, *Phys. Rev. B* 82 (2010), 155305, <https://doi.org/10.1103/PhysRevB.82.155305>.
- [22] D.C. Tripathi, Y.N. Mohapatra, Ideal organic homojunction diode obtained using controlled alignment of localized density of states across doped/undoped interface, *Org. Electron.* 13 (2012) 1680–1685, <https://doi.org/10.1016/j.orgel.2012.04.034>.
- [23] U. Würfel, D. Neher, A. Spies, S. Albrecht, Impact of charge transport on current-voltage characteristics and power-conversion efficiency of organic solar cells, *Nat. Commun.* 6 (2015) 6951, <https://doi.org/10.1038/ncomms7951>.
- [24] C.M. Proctor, T.-Q. Nguyen, Effect of leakage current and shunt resistance on the light intensity dependence of organic solar cells, *Appl. Phys. Lett.* 106 (2015), 083301, <https://doi.org/10.1063/1.4913589>.
- [25] X. Wang, J. Huang, J. Li, J. Yu, Effect of organic electron blocking layers on the performance of organic photodetectors with high ultraviolet detectivity, *J. Phys. D: Appl. Phys.* 49 (2016), 075102, <https://doi.org/10.1088/0022-3727/49/7/075102>.
- [26] V.V. Brus, H.K. Lee, C.M. Proctor, M. Ford, X. Liu, M.A. Burgers, J. Lee, G.C. Bazan, T.Q. Nguyen, Balance between light absorption and recombination losses in solution-processed small molecule solar cells with normal or inverted structures, *Adv. Energy Mater.* 8 (2018) 1–7, <https://doi.org/10.1002/aenm.201801807>.
- [27] J. Bisquert, G. Garcia-Belmonte, A. Munar, M. Sessolo, A. Soriano, H.J. Bolink, Band unpinning and photovoltaic model for P3HT:PCBM organic bulk heterojunctions under illumination, *Chem. Phys. Lett.* 465 (2008) 57–62, <https://doi.org/10.1016/j.cplett.2008.09.035>.
- [28] S.L.M. van Mensfoort, R. Coehoorn, Determination of injection barriers in organic semiconductor devices from capacitance measurements, *Phys. Rev. Lett.* 100 (2008), 086802, <https://doi.org/10.1103/PhysRevLett.100.086802>.
- [29] M. Mingebach, C. Deibel, V. Dyakonov, Built-in potential and validity of the Mott-Schottky analysis in organic bulk heterojunction solar cells, *Phys. Rev. B* 84 (2011), 153201, <https://doi.org/10.1103/PhysRevB.84.153201>.
- [30] D.C. Tripathi, A.K. Tripathi, Y.N. Mohapatra, Mobility determination using frequency dependence of imaginary part of impedance ($\text{Im } Z$) for organic and polymeric thin films, *Appl. Phys. Lett.* 98 (2011), 033304, <https://doi.org/10.1063/1.3544935>.
- [31] T. Kirchartz, W. Gong, S.A. Hawks, T. Agostinelli, R.C.I. MacKenzie, Y. Yang, J. Nelson, Sensitivity of the mott-schottky analysis in organic solar cells, *J. Phys. Chem. C* 116 (2012) 7672–7680, <https://doi.org/10.1021/jp300397f>.
- [32] D.C. Tripathi, Y.N. Mohapatra, Diffusive capacitance in space charge limited organic diodes: analysis of peak in capacitance-voltage characteristics, *Appl. Phys. Lett.* 102 (2013), 253303, <https://doi.org/10.1063/1.4812487>.
- [33] Q. Niu, N.I. Craciun, G.-J.A.H. Wetzelaer, P.W.M. Blom, Origin of negative capacitance in bipolar organic diodes, *Phys. Rev. Lett.* 120 (2018), 116602, <https://doi.org/10.1103/PhysRevLett.120.116602>.
- [34] E. Von Hauff, Impedance spectroscopy for emerging photovoltaics, *J. Phys. Chem. C* 123 (2019) 11329–11346, <https://doi.org/10.1021/acs.jpcc.9b00892>.
- [35] D.C. Tripathi, K.S. Rao, Y.N. Mohapatra, Impact of defect states on the capacitance voltage characteristics of space charge limited organic diodes, and determination of defect states, *J. Mater. Chem. C* 9 (2021) 4903–4909, <https://doi.org/10.1039/D1TC00027F>.
- [36] S. Banerjee, S.K. Gupta, A. Singh, A. Garg, Buffer layers in inverted organic solar cells and their impact on the interface and device characteristics: an experimental and modeling analysis, *Org. Electron.* 37 (2016) 228–238, <https://doi.org/10.1016/j.orgel.2016.06.031>.
- [37] S.K. Gupta, L.S. Pali, A. Garg, Impedance spectroscopy on degradation analysis of polymer/fullerene solar cells, *Sol. Energy* 178 (2019) 133–141, <https://doi.org/10.1016/j.solener.2018.12.024>.
- [38] W.C. Germs, S.L.M. van Mensfoort, R.J. de Vries, R. Coehoorn, Effects of energetic disorder on the low-frequency differential capacitance of organic light emitting diodes, *J. Appl. Phys.* 111 (2012), 074506, <https://doi.org/10.1063/1.3701575>.
- [39] J.C. Nolasco, A. Sánchez-Díaz, R. Cabr, J. Ferr-Borrull, L.F. Marsal, E. Palomares, J. Pallares, Relation between the barrier interface and the built-in potential in pentacene/C60 solar cell, *Appl. Phys. Lett.* 97 (2010) 1–4, <https://doi.org/10.1063/1.3456393>.
- [40] U.K. Verma, S. Kumar, Y.N. Mohapatra, Measurement of contact surface photo-voltage from forward bias C-V characteristics of P3HT:PCBM based BHJ solar cells, *Sol. Energy Mater. Sol. Cells* 172 (2017) 25–33, <https://doi.org/10.1016/j.solmat.2017.07.010>.
- [41] S.K. Gupta, D.C. Tripathi, A. Garg, S.K. Pathak, Determination of defect states and surface photovoltage in PTB7:PC71BM based bulk heterojunction solar cells, *Sol. Energy Mater. Sol. Cells* 224 (2021), 110994, <https://doi.org/10.1016/j.solmat.2021.110994>.
- [42] B. Ray, A.G. Baradwaj, B.W. Boudouris, M.A. Alam, Defect characterization in organic semiconductors by forward bias capacitance-voltage (FB-CV) analysis, *J. Phys. Chem. C* 118 (2014) 17461–17466, <https://doi.org/10.1021/jp505500r>.
- [43] A. Sharma, P. Kumar, B. Singh, S.R. Chaudhuri, S. Ghosh, Capacitance-voltage characteristics of organic Schottky diode with and without deep traps, *Appl. Phys. Lett.* 99 (2011), 023301, <https://doi.org/10.1063/1.3607955>.
- [44] T.-Y. Chu, O.-K. Song, Thickness dependence of the trap states in organic thin film of N,N'-bis(naphthalen-1-yl)-N,N'-bis(phenyl) benzidine, *Appl. Phys. Lett.* 91 (2007), 073508, <https://doi.org/10.1063/1.2771536>.
- [45] J. Bisquert, G. Garcia-Belmonte, A. Pitarch, H.J. Bolink, Negative capacitance caused by electron injection through interfacial states in organic light-emitting diodes, *Chem. Phys. Lett.* 422 (2006) 184–191, <https://doi.org/10.1016/j.cplett.2006.02.060>.
- [46] E. Ehrenfreund, C. Lungenschmied, G. Dennler, H. Neugebauer, N.S. Sariciftci, Negative capacitance in organic semiconductor devices: bipolar injection and charge recombination mechanism, *Appl. Phys. Lett.* 91 (2007), 012112, <https://doi.org/10.1063/1.2752024>.
- [47] T.K. Djidjou, T. Basel, A. Rogachev, Admittance spectroscopy study of polymer diodes in small magnetic fields, *J. Appl. Phys.* 112 (2012), 024511, <https://doi.org/10.1063/1.4737773>.
- [48] S. Kumar, U.K. Verma, Y.N. Mohapatra, Negative contribution to the reverse bias capacitance of organic diodes due to field dependent mobility: determination of barrier height and transport parameters, *J. Appl. Phys.* 124 (2018), 035501, <https://doi.org/10.1063/1.5026479>.
- [49] H.H.P. Gommans, M. Kemerink, R.A.J. Janssen, Negative capacitances in low-mobility solids, *Phys. Rev. B* 72 (2005), 235204, <https://doi.org/10.1103/PhysRevB.72.235204>.
- [50] E. Knapp, B. Ruhstaller, Analysis of negative capacitance and self-heating in organic semiconductor devices, *J. Appl. Phys.* 117 (2015), 135501, <https://doi.org/10.1063/1.4916981>.
- [51] L.S.C. Pingree, M.T. Russell, T.J. Marks, M.C. Hersam, Field dependent negative capacitance in small-molecule organic light-emitting diodes, *J. Appl. Phys.* 100 (2006), 044502, <https://doi.org/10.1063/1.2234544>.
- [52] S.K. Gupta, S. Banerjee, A. Singh, L.S. Pali, A. Garg, Modeling of degradation in normal and inverted OSC devices, *Sol. Energy Mater. Sol. Cells* 191 (2019), <https://doi.org/10.1016/j.solmat.2018.11.039>.
- [53] S.K. Gupta, A. Sharma, S. Banerjee, R. Gahlot, N. Aggarwal, Deepak, A. Garg, Understanding the role of thickness and morphology of the constituent layers on the performance of inverted organic solar cells, *Sol. Energy Mater. Sol. Cells* 116 (2013) 135–143, <https://doi.org/10.1016/j.solmat.2013.03.027>.
- [54] S. Banerjee, S.K. Gupta, A. Singh, A. Garg, Buffer layers in inverted organic solar cells and their impact on the interface and device characteristics: an experimental and modeling analysis, *Org. Electron. Phys., Mater. Appl.* 37 (2016) 228–238, <https://doi.org/10.1016/j.orgel.2016.06.031>.
- [55] W.J. Potscavage, S. Yoo, B. Kippelen, Origin of the open-circuit voltage in multilayer heterojunction organic solar cells, *Appl. Phys. Lett.* 93 (2008), 193308, <https://doi.org/10.1063/1.3027061>.
- [56] H. Shekhar, O. Solomeshch, D. Liraz, N. Tessler, Low dark leakage current in organic planar heterojunction photodiodes, *Appl. Phys. Lett.* 111 (2017), 223301, <https://doi.org/10.1063/1.4996826>.
- [57] S. Dongaonkar, J.D. Servaites, G.M. Ford, S. Loser, J. Moore, R.M. Gelfand, H. Mohseni, H.W. Hillhouse, R. Agrawal, M.A. Ratner, T.J. Marks, M.S. Lundstrom, M.A. Alam, Universality of non-Ohmic shunt leakage in thin-film solar cells, *J. Appl. Phys.* 108 (2010), 124509, <https://doi.org/10.1063/1.3518509>.
- [58] S. Yoon, J. Cho, K.M. Sim, J. Ha, D.S. Chung, Low dark current inverted organic photodiodes using anionic polyelectrolyte as a cathode interlayer, *Appl. Phys. Lett.* 110 (2017), 083301, <https://doi.org/10.1063/1.4977025>.

- [59] G.A.H. Wetzelaer, Analytical description of the current-voltage relationship in organic-semiconductor diodes, *AIP Adv.* 8 (2018), 035320, <https://doi.org/10.1063/1.5021989>.
- [60] www.synopsys.com, Sentaurus Device User Guide J-2014.09, (2014) 1488.
- [61] R.M. Hill, Poole-Frenkel conduction in amorphous solids, *Philos. Mag.* 23 (1971) 59–86, <https://doi.org/10.1080/14786437108216365>.
- [62] C. Deibel, A. Wagenpfahl, V. Dyakonov, Origin of reduced polaron recombination in organic semiconductor devices, *Phys. Rev. B* 80 (2009), 075203, <https://doi.org/10.1103/PhysRevB.80.075203>.
- [63] G.J. Adriaenssens, V.I. Arkhipov, Non-Langevin recombination in disordered materials with random potential distributions, *Solid State Commun.* 103 (1997) 541–543, [https://doi.org/10.1016/S0038-1098\(97\)00233-0](https://doi.org/10.1016/S0038-1098(97)00233-0).
- [64] G. Juška, K. Arlauskas, J. Stuchlik, R. Österbacka, Non-Langevin bimolecular recombination in low-mobility materials, *J. Non Cryst. Solids* 352 (2006) 1167–1171, <https://doi.org/10.1016/j.jnoncrysol.2005.11.099>.